

Juwon Kim

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Research Interests

Operating System, Filesystems, Flash storage, Device firmware (Flash Translation Layers), Garbage collection, and Learned index.

Education

Ph.D. in Electrical Engineering *Mar. 2023 – Aug. 2026*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Advisor: Prof. Youjip Won

M.S. in Electrical Engineering *Mar. 2021 – Feb. 2023*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Thesis: *IPLFS: Log-Structured Filesystem without Garbage Collection*
- Advisor: Prof. Youjip Won

B.S. in Electrical Engineering *Mar. 2016 – Feb. 2021*
Korea Advanced Institute of Science and Technology (KAIST) Daejeon, Korea

- Individually Designed Major in Computer Science and Physics

Publications

- [1] **[ACM SOSP '26]** Juwon Kim, Hwanseok Yoo, Seung Won Yoo, and Youjip Won, “GAZE: Rearchitecting Log-Structured Filesystem for Garbage Collection on Zoned Namespace Flash Storage”, to appear in *Proc. ACM Symposium on Operating Systems Principles (SOSP)*, Prague, Czechia, Sep.–Oct. 2026.
- [2] **[IEEE NVMSA '26]** Jihoon Seok, Juwon Kim, Youjip Won, “LeaST: Learned Sector Translation for Skipped Physical Page Address”, in *Proc. IEEE Non-Volatile Memory Systems and Applications Symposium (NVMSA)*, Aug. 2026.
- [3] **[USENIX FAST '25]** Juwon Kim, Seungjae Lee, Joontaek Oh, Dongkun Shin, Youjip Won, “D2FS: Device-Driven Filesystem Garbage Collection”, in *Proc. USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, USA, Feb. 2025.
- [4] **[USENIX FAST '25]** Jieun Kim, Joontaek Oh, Juwon Kim, Seung Won Yoo, Youjip Won, “OPIMQ: Order Preserving IO Stack for Multi-Queue Block Device”, in *Proc. USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, USA, Feb. 2025.
- [5] **[ACM SAC '25]** Juwon Kim, Dongeon Kim, Seung Won Yoo, Myoengin Cheon, Joontaek Oh, Youjip Won, “lwFSCK: Light-weight Filesystem Check”, in *Proc. ACM Symposium on Applied Computing (SAC)*, Catania, Italy, Mar. 2025.
- [6] **[USENIX ATC '22]** Juwon Kim, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection”, in *Proc. USENIX Annual Technical Conference (ATC)*, San Diego, USA, Jul. 2022.

Posters and Workshops

- [1] **[ACM SOSP '25]** Juwon Kim, Hwanseok Yoo, Youjip Won, “Accelerating Filesystem Garbage Collection for Zoned Flash Storage,” in *Proc. ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025.
- [2] **[ACM SOSP '25]** Hwanseok Yoo, Juwon Kim, Youjip Won, “Manycore Scalability of Mod-

ern Log-Structured Filesystem: Revisited,” in *Proc. ACM Symposium on Operating Systems Principles (SOSP)*, Seoul, Republic of Korea, Oct. 2025.

- [3] [USENIX FAST '22] **Juwon Kim**, Minsu Jang, Danish Muhammad Teeshen, Joontaek Oh, Youjip Won, “IPLFS: Log-Structured File System without Garbage Collection,” in *Proc. USENIX Conference on File and Storage Technologies (FAST)*, Santa Clara, USA, Feb. 2022.

Patents

- [1] Youjip Won and **Juwon Kim**, “Method and Device for Managing Host Area,” Korean Patent Application No. 10-2025-0174419, filed Nov. 2025.
- [2] **Juwon Kim** and Youjip Won, “Storage Device Performing Garbage Collection,” Korean Patent Application No. 10-2025-0090393, filed Jul. 2025. (*Joint filing: SK Hynix & KAIST*)

Awards & Honors

- Student Grant (Full), USENIX FAST, USA Feb. 2025
- NVRAMOS 2023 OpenSSD Best Practice Award, Gold Prize, KIISE, Korea Nov. 2023
- Student Grant (Full), USENIX FAST, USA Feb. 2022

Research Projects

Order Preserving Distributed Storage System Jun. 2025 – Present
NRF, Korea

Optimizing Filesystem Garbage Collection for Zoned Flash Storage Sep. 2024 – Present
SNU-SKH Solution Research Center, SK Hynix, Korea

Device-Driven Filesystem Garbage Collection Sep. 2023 – Aug. 2024
SNU-SKH Solution Research Center, SK Hynix, Korea

Collection and Integrated Analysis of Automotive Inter/Intra System Artifacts Jan. 2024 – Feb. 2025
IITP, Korea

Future Scalable OS: Log-Structured Filesystem without Garbage Collection Jan. 2021 – May 2023
IITP, Korea

Teaching Experience

- Head TA, Operating Systems and System Programming for EE (EE415), KAIST Sep.–Dec. 2025
- TA, Special Topics in Computer Engineering (EE817), KAIST Feb.–Jun. 2025
- TA, Operating Systems and System Programming for EE (EE415), KAIST Sep.–Dec. 2024
- Head TA, Programming Structure for EE (EE209), KAIST Feb.–Jun. 2024
- TA, Programming Structure for EE (EE209), KAIST Sep.–Dec. 2023
- TA, Programming Structure for EE (EE209), KAIST Feb.–Jun. 2023

Service

- Artifact Evaluation Committee, USENIX FAST '27 Spring Cycle Jun. 2026 – Jul. 2026
- Program Chair Assistant, USENIX FAST '24 Sep. 2023 – Sep. 2024

Technical Skills

Systems: F2FS, flash translation layer, NVMe driver, block I/O layer, ZNS, Page Management Subsystem

Languages: C, Python, C++, SQL, Verilog

Tools: git, vim, gnuplot, ctags, cscope, ftrace, blktrace